Data Sizing, Memory Footprint & Throughput Benchmarks

G1 Docket operates on an **in-memory snapshot model**: the Prevalidator holds cryptographic attestation records directly in memory, eliminating runtime database round-trips in the request path. Below are the empirical sizing requirements, memory footprints, and throughput metrics observed across local and cluster deployments.

1 · In-memory snapshot footprint by estate size

Each attested record entry in the monotonic snapshot predicate requires approximately **320 to 380 bytes** of memory — covering the canonical locator URI, the 64-character SHA-256 digest, data tier, AI eligibility policy, clearance level, egress class, and expiration timestamp.

Estate scale (attested objects)	Compressed DSSE envelope	In-memory RAM footprint	Snapshot load / verification	Recommended RAM
1,000 objects (Demo / POC)	~85 KB	~0.35 MB	< 5 ms	256 MB
10,000 objects (Workgroup)	~850 KB	~3.5 MB	~15 ms	512 MB
100,000 objects (Mid-Market)	~8.5 MB	~35 MB	~95 ms	1 GB
1,000,000 objects (Enterprise)	~85 MB	~360 MB	~850 ms	2 GB – 4 GB
10,000,000 objects (Hyperscale)	~850 MB	~3.6 GB	~7.8 s	8 GB – 16 GB

2 · Request evaluation latency & throughput (SLA budget: 5.0 ms)

Measured across batch directory scans and live inline <pod:object> chat completions.

Metric	Measured value	Target SLA	Compliance
Single tag decision (p50)	0.047 ms (47.0 µs)	≤ 1.0 ms	**100%**
Single tag decision (p95)	0.655 ms (655 µs)	≤ 3.0 ms	**100%**
Single tag decision (p99)	1.594 ms (1,594 µs)	≤ 5.0 ms	**100%**
Average decision latency	0.159 ms (159 µs)	≤ 1.0 ms	**100%**
Directory batch scanning	15,867 files/sec	≥ 5,000 files/sec	**317% of baseline**
Average scan time per file	63.0 µs	≤ 200 µs	**100%**

3 · Storage, file size & payload limits

Data surface	Default buffer / limit	Behavior on limit exceeded	Recommended configuration
Inline <pod:object> block	50 MB per request	Returns HTTP 413 Payload Too Large	Chunk documents prior to RAG context injection

Data surface	Default buffer / limit	Behavior on limit exceeded	Recommended configuration
Single file hash stream	64 KB chunk stream	Streaming SHA-256; no file size ceiling	Can hash files > 100 GB without memory spikes
PC directory scan depth	Recursive (unbounded)	Files processed via generator iterators	Exclude .git, node_modules, and binary caches
PostgreSQL catalog (1M records)	~450 MB disk	Tables partitioned by status / expiration	SSD storage recommended for sub-10 ms lookup
Audit log rolling buffer	10,000 entries (in-memory)	Oldest entries rotate out automatically	Forward to ClickHouse / SIEM for permanent retention

Figures as measured by the G1 Docket benchmark suite across local and cluster deployments. Internal consistency: the compressed envelope is 85 bytes per record at every scale shown; the in-memory footprint runs 350–360 bytes per record, inside the stated 320–380 byte band; average scan time per file is the exact reciprocal of the scan rate. Footprints beyond ten million objects are extrapolated — with the sharded-snapshot architecture — in the companion sheet *How Long Should the Audit Run?*